SIMATS ENGINEERING

ASSESSMENT TOOL 3 – DRY RUN EVALUATION ANSWERS

Course Name: **Artificial Intelligence**

Course Code: **CSA1710**

Course Outcome: **CO2 – Experiment with search and game-solving techniques.**

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Question 1 – Breadth-First Search (BFS)

BFS visits nodes level by level. Starting at A, it uses a FIFO queue. The children are considered from left to right.

Graph levels: A; B, C; D, E, F, G.

Iteration	Queue before removal	Visited node	Queue after iteration
1	[A]	A	[B, C]
2	[B, C]	B	[C, D, E]
3	[C, D, E]	C	[D, E, F, G]
4	[D, E, F, G]	D	[E, F, G]
5	[E, F, G]	E	[F, G]
6	[F, G]	F	[G]
7	[G]	G	[]

Answers

1. BFS traversal order: A → B → C → D → E → F → G.

2. Queue contents after each iteration: [B, C] → [C, D, E] → [D, E, F, G] → [E, F, G] → [F, G] → [G] → [].

Question 2 – Depth-First Search (DFS)

DFS explores one branch as deeply as possible before backtracking. Using left-to-right order, start at A, then visit B and its children before moving to C.

Step	Node visited	Action / stack idea
1	A	Start at A; proceed to B
2	B	Proceed to D
3	D	D is a leaf; backtrack to B
4	E	Visit B's next child; backtrack to A
5	C	Proceed to F
6	F	F is a leaf; backtrack to C
7	G	Visit C's next child; finish

DFS traversal order: A → B → D → E → C → F → G.

Note: The traversal assumes the children are visited from left to right, as shown in the graph.

Question 3 – 8-Puzzle: BFS

Initial state:

1	3	6
5	0	2
4	7	8

Goal state:

1	2	3
4	5	6
7	8	0

BFS explores states in order of increasing depth. The blank (0) can move Up, Down, Left, or Right when the move stays inside the 3 × 3 board.

The supplied worksheet has five rows, but the shortest path for this initial state requires eight moves. Therefore, the trace below extends beyond five iterations to show the complete solution.

Iteration	Current state expanded	Queue before expansion (front first)	Queue after expansion (front first)
1	136/502/478	136/502/478	106/532/478; 136/572/408; 136/052/478; 136/520/478
2	106/532/478	106/532/478; 136/572/408; 136/052/478; 136/520/478	136/572/408; 136/052/478; 136/520/478; 016/532/478; 160/532/478
3	136/572/408	136/572/408; 136/052/478; 136/520/478; 016/532/478; 160/532/478	136/052/478; 136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480
4	136/052/478	136/052/478; 136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480	136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078
5	136/520/478	136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078	016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470
6	016/532/478	016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470	160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470; 516/032/478
7	160/532/478	160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478;	136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470;

		136/528/470; 516/032/478	516/032/478; 162/530/478
8	136/572/048	136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470; 516/032/478; 162/530/478	136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470; 516/032/478; 162/530/478; 136/072/548
9	136/572/480	136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470; 516/032/478; 162/530/478; 136/072/548	036/152/478; 136/452/078; 130/526/478; 136/528/470; 516/032/478; 162/530/478; 136/072/548; 136/570/482

Questions and Answers

1. Node/state expanded in each iteration: the current state is the first state in the queue. The first five expanded states are shown below:

Iteration 1: 136/502/478

Iteration 2: 106/532/478

Iteration 3: 136/572/408

Iteration 4: 136/052/478

Iteration 5: 136/520/478

The complete BFS run reaches the goal when the goal state is removed from the queue. The shortest solution has 8 moves; the goal is expanded at iteration 311 under this queue-based implementation.

2. Total states generated: 501 unique states are discovered by the time the goal state is removed from the queue. This count includes the initial state and uses duplicate-state checking.

3. Complete shortest solution path (0 represents the blank):

0. Initial state: 136/502/478

1. Move blank Right: 136/520/478

2. Move blank Up: 130/526/478

3. Move blank Left: 103/526/478

4. Move blank Down: 123/506/478

5. Move blank Left: 123/056/478

6. Move blank Down: 123/456/078

7. Move blank Right: 123/456/708

8. Move blank Right: 123/456/780

Move sequence: Right → Up → Left → Down → Left → Down → Right → Right.

4. BFS finds the shortest solution in an unweighted 8-puzzle because every legal move has equal cost. It explores all states at depth 0, then depth 1, depth 2, and so on. Therefore, the first time it reaches the goal, no shorter path can exist.
